



Introduction to the Program Design and Programming

ICT 1305

CHAPTER 05

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What is Pseudocode

- Flowcharts were the first design tool to be widely used, but unfortunately they do not reflect some of the concepts of structured programming very well.
- Pseudocode, on the other hand, is a newer tool and has features that make it more reflective of the structured concepts.
- Pseudocode is a technique used to describe the distinct steps of an algorithm in a manner that's easy to understand for anyone with basic programming knowledge.
- The drawback is that the narrative presentation is not as easy to understand and/or follow.



Pseudocode

- Pseudocode is represent six programming constructs:
 - SEQUENCE
 - CASE
 - WHILE
 - REPEAT-UNTIL
 - FOR
 - IF-THEN-ELSE
- These constructs are used to describe the control flow of the algorithm.

PSEUDOCODE CONSTRUCTS

SEQUENCE

Input: READ, OBTAIN, GET

Output: PRINT, DISPLAY, SHOW

Compute: COMPUTE,
CALCULATE, DETERMINE

Initialize: SET, INIT

Add: INCREMENT, BUMP

Sub: DECREMENT

FOR

FOR iteration bounds

sequence

ENDFOR

WHILE

WHILE condition

sequence

ENDWHILE

CASE

CASE expression OF

condition 1: sequence 1

condition 2: sequence 2

...

condition n: sequence n

OTHERS:

default sequence

ENDCASE

REPEAT-UNTIL

REPEAT

sequence

UNTIL condition

IF-THEN-ELSE

IF condition THEN

sequence 1

ELSE

sequence 2

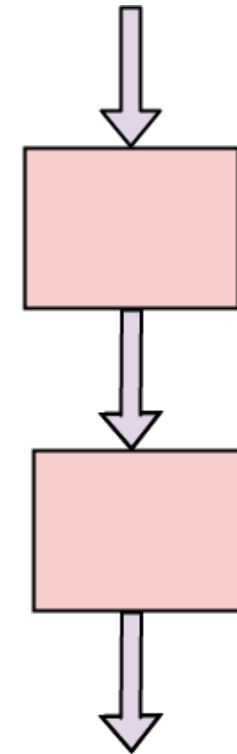
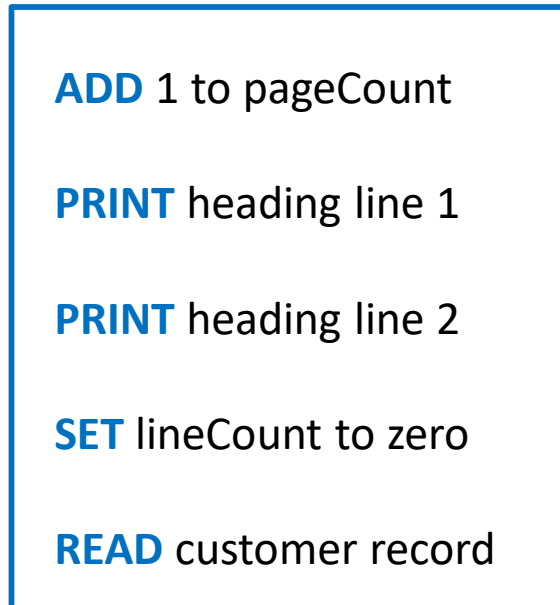
ENDIF

The Structure Theorem

- It is possible to write any computer program by using only three basic control structures that are easily represented in pseudocode:
 - Sequence
 - Selection
 - Repetition

Sequence

- Is the straightforward execution of one processing step after another.



Selection

- Presentation of condition and the choice between two actions.

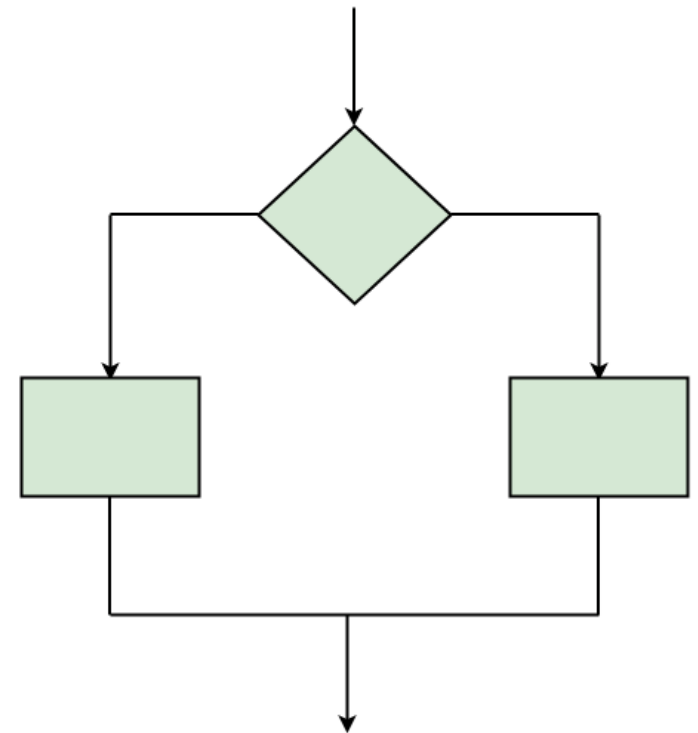
IF condition p is true **THEN**

statement(s) in true case

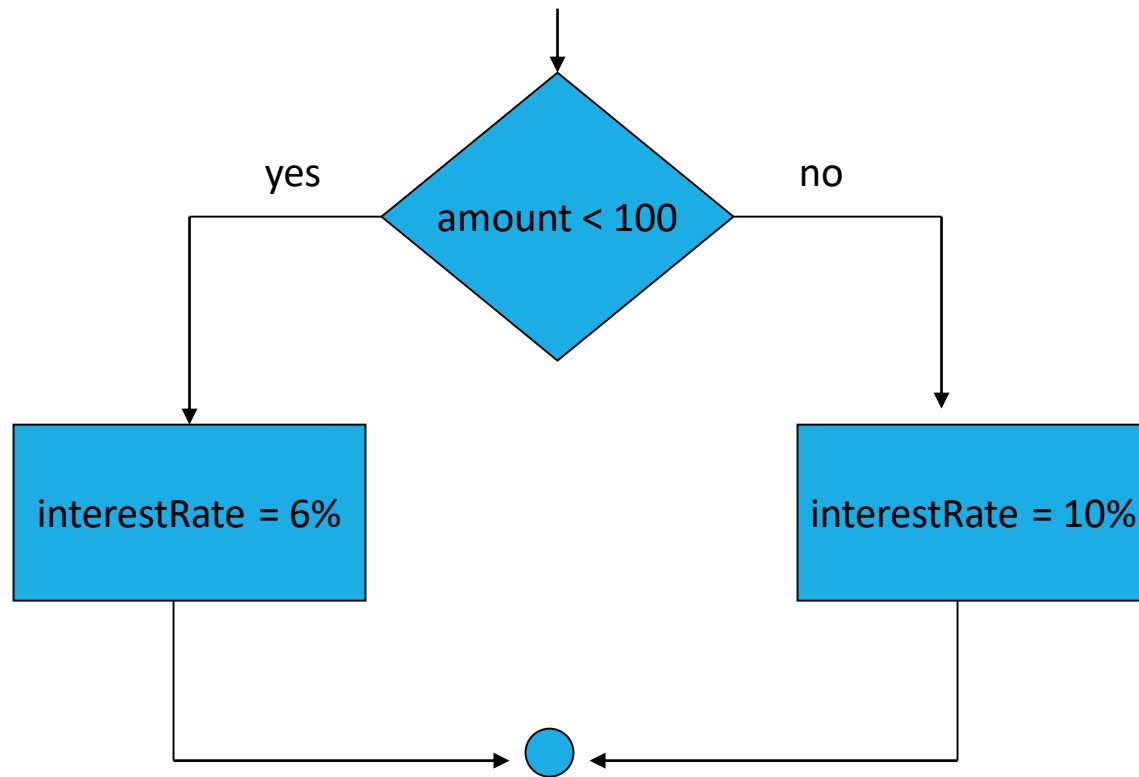
ELSE

statement(s) in false case

ENDIF



Selection



IF amount < 100

interestRate = 6%

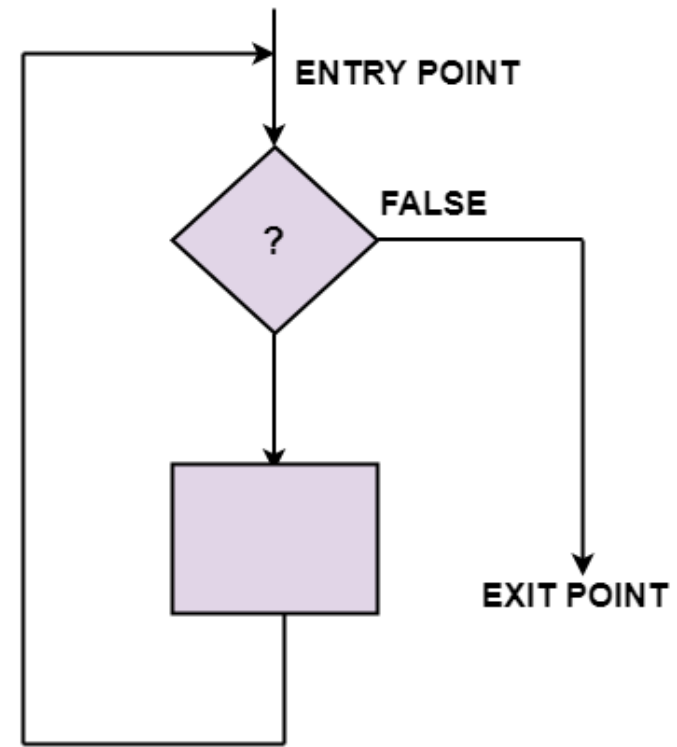
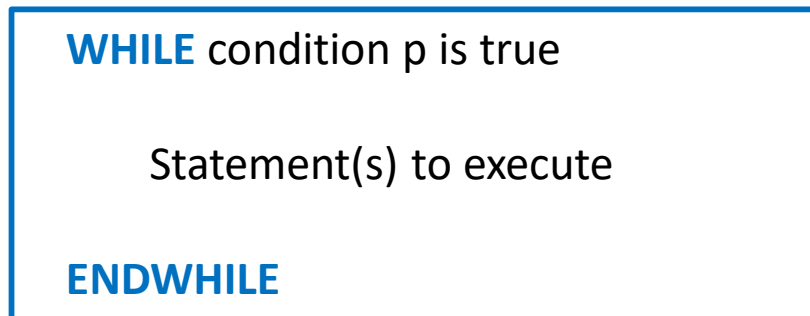
ELSE

interestRate = 10%

ENDIF

Repetition

- The presentation of a set of instructions to be performed repeatedly, as long as a condition is true



Repetition

Example:

```
SET student_total to 0

WHILE student_total < 50

    READ student record

    PRINT student name and address

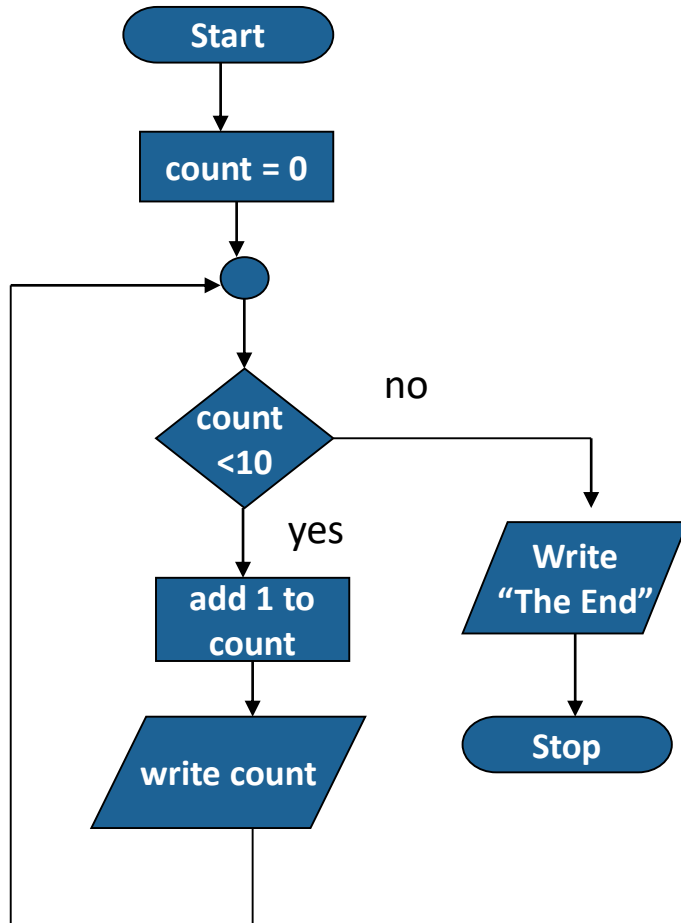
    ADD 1 to student_total

ENDWHILE
```

The Looping Structure

- In flowcharting one of the more confusing things is to separate selection from looping.
- This is because each structure use the diamond as their control symbol.
- In pseudocode avoid this by using specific keywords to designate looping,
 - WHILE/ENDWHILE
 - REPEAT/UNTIL

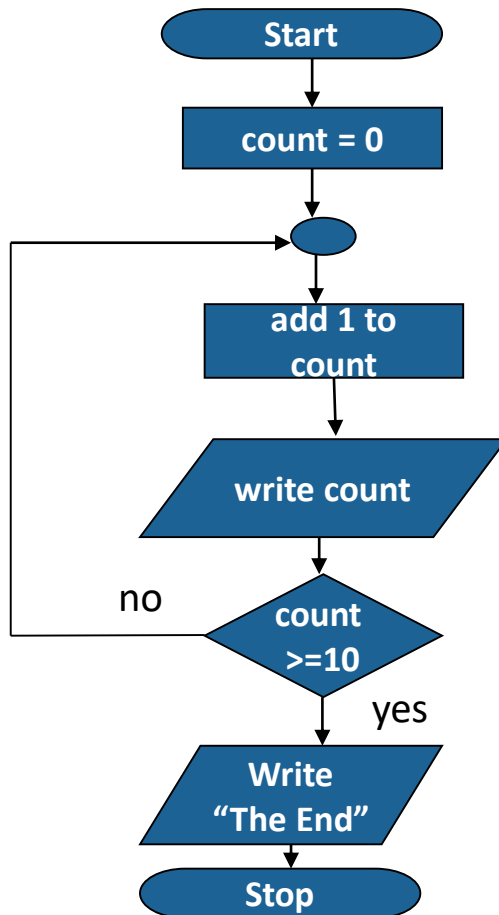
WHILE / ENDWHILE



```
count = 0  
  
WHILE count < 10  
    ADD 1 to count  
    WRITE count  
  
ENDWHILE  
  
WRITE "The End"
```

```
Mainline  
count = 0  
WHILE count < 10  
    DO Process  
ENDWHILE  
WRITE "The End"  
  
Process  
ADD 1 to count  
WRITE count
```

REPEAT / UNTIL



count = 0

REPEAT

ADD 1 to count

WRITE count

UNTIL count >= 10

WRITE "The End"

Mainline

count = 0

REPEAT

DO Process

UNTIL count >= 10

WRITE "The End"

Process

ADD 1 to count

WRITE count

Rules for Pseudocode

1. Capitalize initial keywords
2. Write only one statement per line
3. Indent to show hierarchy
4. End multiline sections
5. Keep statements language independent
6. Use the naming domain of the problem

Capitalize Initial Keyword

- In the example below note the words: READ and WRITE. These are just a few of the keywords to use, others include:

READ, WRITE, IF, ELSE, ENDIF, WHILE, ENDWHILE

```
READ name, hoursWorked, payRate
```

```
gross = hoursWorked * payRate
```

```
WRITE name, hoursWorked, gross
```

One statement per line

- Each statement in pseudocode should express just one action for the computer.
- If the task list is properly drawn, then in most cases each task will correspond to one line of pseudocode.

Task List

1. Read name, hours worked, rate of pay
2. Perform calculations
3. gross = hours worked * rate of pay
4. Write name, hours worked, gross



Pseudocode

```
READ name, hoursWorked, payRate  
gross = hoursWorked * payRate  
WRITE name, hoursWorked, gross
```


Indent to Show Hierarchy

- Each design structure uses a particular indentation pattern
- Sequence:
 - Keep statements in sequence all starting in the same column
- Selection:
 - Indent statements that fall inside selection structure, but not the keywords that form the selection

Indent to Show Hierarchy

- Looping: Indent statements that fall inside the loop, but not keywords that form the loop

TASK LIST:

1. Read name, hourly rate, hours worked, deduction rate
2. Compute gross, deduction, net pay
3. Is gross ≥ 100 ?
4. YES: calculate deduction
5. NO: no deduction
6. Write name, gross, deduction, net pay



PSEUDOCODE:

```
READ name, hourlyRate, hoursWorked
grossPay = hourlyRate * hoursWorked
IF grossPay  $\geq$  100
    deduction = grossPay * deductionRate
ELSE
    deduction = 0
ENDIF
netPay = grossPay – deduction
WRITE name, grossPay, deduction, netPay
```

End Multiline Structures

- See the IF/ELSE/ENDIF as constructed, the ENDIF is in line with the IF.
- The same applies for WHILE/ENDWHILE etc...

```
READ name, grossPay, taxes
IF taxes > 0
    net = grossPay - taxes
ELSE
    net = grossPay
ENDIF
WRITE name, net
```

Language Independence

- Resist the urge to write in whatever language you are most comfortable with, in the long run you will save time.
- Remember you are describing a logic plan to develop a program, you are not programming!



Six Basic Computer Operations

1. A computer can receive information
2. A computer can output information
3. A computer can perform arithmetic operations
4. A computer can assign a value to a variable
5. A computer can compare two variables and select one of two alternate actions
6. A computer can repeat a group of actions

1. Computer can receive information

Verb used:

- **READ** used when the algorithm is to receive the input from a record on a file
- **GET / INPUT** used when the algorithm is to receive input from the keyboard.

READ *student_name*

GET *system_date*

READ *number_1, number_2*

GET *tax_code*

INPUT *marks*

2. Computer can output information

Verb used:

- **PRINT** used when the output is to be sent to the printer.
- **WRITE** used when the output is to be written to a file.
- **PUT, OUTPUT, DISPLAY** used when the output is to be written to the screen.
- **PROMPT** required before an input instruction Get, causes the message to be sent to the screen which requires the user responds, usually by providing input.

2. Computer can output information

PRINT `Program Completed`

WRITE *customer_record* to master file

PUT *name, address and postcode*

OUTPUT *total_tax*

DISPLAY `End of data`

PROMPT for *student_mark*

GET *student_mark*

3. Computer can perform arithmetic operations

Symbols used:

+	Add
-	Subtract
*	Multiply
/	Divide
()	Parentheses

Verb used:

- Compute
- Calculate

ADD *number to total*

Total = total + number

DIVIDE *total_marks by student_count*

*Sales_tax = cost_price * 0.10*

COMPUTE *C = (F - 32) * 5/9*

4. Computer can assign a value to a variable

Three cases :

- To give data an initial value in pseudocode, the verbs **INITIALIZE** or **SET** are used
- To assign a value as a result of some processing, the symbols '=' or '<-' are written
- To keep a variable for later use, the verbs **SAVE** or **STORE** are used.

4. Computer can assign a value to a variable

INITIALIZE *total_price* to zero

SET *student_count* to 0

Total_price = *cost_price* + *sales_tax*

Total_price <- *cost_price* + *sales_tax*

STORE *customer_num* in *last_customer_num*

5. Computer can compare two variables and select one of two alternate actions

- To represent this operation in Pseudocode, special keywords are used: **IF**, **THEN** and **ELSE**.

```
IF student is part_time THEN  
    ADD 1 to part_time count  
  
ELSE  
    ADD 1 to full_time count  
  
ENDIF
```

6. Computer can repeat a group of actions

Keyword used:

- DO WHILE, END DO
- WHILE, END WHILE

```
DO WHILE student_total < 50
```

```
    READ studentRecord
```

```
    PRINT studentName, address to report
```

```
    ADD 1 to student_total
```

```
END DO
```

Why Use Pseudocode?

- Pseudocode Is Easier to Read
- Pseudocode Simplifies Code Construction
- Pseudocode Is a Good Middle Point Between Flowchart and Code
- Pseudocode Is a Helpful Starting Point for Documentation
- Pseudocode Allows for Quick Bug Detection

Rules for Variable Names

- Begin with lowercase letter
- Contain no spaces
- Additional words begin with capital
- Unique names within code
- Consistent use of names

INITIALIZE number

SET firstName to “Kamal”

INITIALIZE offer_rate

Desk Checking a Pseudocode

- Choose valid simple input test case (2-3 enough)
- Establish what the expected result should be.
- Make a table of relevant variable names
- Checking the test case line by line, step by step
- Repeat process for other test case
- Check if expected result matches with actual result

Advantages of Pseudocode

- Easy to write - Writing of pseudocode involves much less time and effort than drawing an equivalent flowchart.
- Easy to convert - Converting a pseudo code to a programming language is much more easier as compared to converting a flowchart.
- Simple to change - It is easier to modify the pseudocode of a program logic when program modifications are necessary.

Disadvantages of Pseudocode

- In case of pseudo code, a graphic representation of program logic is not available.
- There are no standard rules to follow in using pseudocode.
- Different programmers use their own style of writing pseudocode.
- Communication problems occur due to lack of standardization.
- For a beginner, it is more difficult to follow the logic or write the pseudo code, as compared to flowcharting.

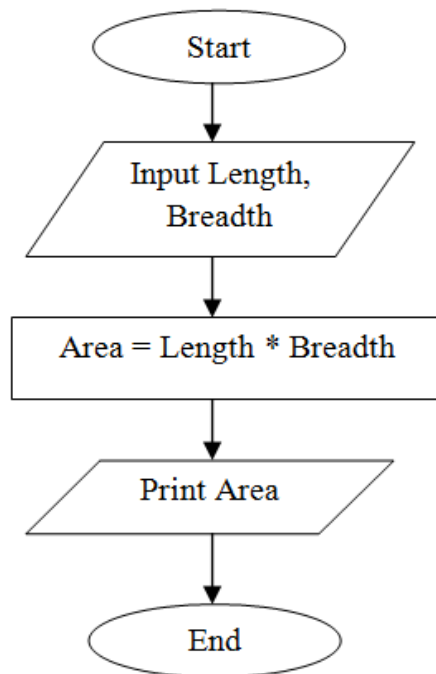
Difference between Algorithm and Pseudocode

Algorithm	Pseudocode
An Algorithm is used to provide a solution to a particular problem in form of a well-defined step-based form.	A Pseudocode is a step-by-step description of an algorithm in code-like structure using plain English text.
An algorithm only uses simple English words	Pseudocode also uses reserved keywords like if-else, for, while, etc.
These are a sequence of steps of a solution to a problem	These are fake codes as the word pseudo means fake, using code like structure and plain English text
There are no rules to writing algorithms	There are certain rules for writing pseudocode
Algorithms can be considered pseudocode	Pseudocode cannot be considered an algorithm
It is difficult to understand and interpret	It is easy to understand and interpret

Examples of Pseudocode

Find the area of rectangle

Flow Chart



Algorithm

Step 1: Start

Step 2: get length,breadth values

Step 3: Calculate area = length*breadth

Step 4: Display area

Step 5: Stop

Pseudocode

BEGIN

READ Length, Breadth

Area = Length * Breadth

DISPLAY Area

END

Examples of Pseudocode

Pseudocode to find the maximum number in a list of numbers.

```
BEGIN  
INPUT list of numbers  
SET max to first number in the list  
FOR each number in the list  
    IF number > max THEN  
        SET max to number  
    ENDIF  
ENDFOR  
OUTPUT max  
END
```

Examples of Pseudocode

Pseudocode that will take two numbers as input, calculates their sum and displays output.

BEGIN

WRITE "Please enter two numbers to add"

READ num1

READ num2

Sum = num1 + num2

WRITE Sum

or

WRITE "The Answer is: ", Sum

END

Examples of Pseudocode

Pseudocode that will take marks of physics, chemistry and math as input, calculates the average and displays output.

BEGIN

WRITE "please enter marks of physics"

INPUT Phy_marks

WRITE "please enter marks of chemistry"

INPUT Chem_marks

WRITE "please enter marks of maths"

INPUT math_marks

$Avg = (Phy_marks + Chem_marks + math_marks) / 3$

DISPLAY Avg

END

Examples of Pseudocode

Pseudo code to calculate sum and average for n numbers.

```
BEGIN  
INITIALIZE sum=0, i=1  
READ n  
FOR i <= n, THEN  
    COMPUTE sum = sum + i  
    CALCULATE i=i+1  
END FOR  
COMPUTE avg = sum/n  
PRINT sum, avg  
END
```


Thank You
